MICROBIOLOGY LABORATORY STUDENT MANUAL

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Microbiology Laboratory Student Manual

This text was based on work by Drew Laux, who made an instructor version of the Northeast Wisconsin Technical College Microbiology Lab manual in 2019.

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A detailed breakdown of this resource's licensing can be found in ***Back Matter/Detailed Licensing***.

CHAPTER OVERVIEW

Introduction

0.1: Course Safety

Safety in the microbiology laboratory is paramount. Students are expected to adhere to all rules to ensure a safe learning environment.

Personal Protective Equipment (PPE) & Attire

- Lab coats and appropriately sized gloves are mandatory and must be worn at all times while in the lab.
- Pants/skirts must cover the entire leg to the ankle; shorts, capris, or excessively ripped jeans are not allowed.
- Closed-toed shoes that cover the entire foot are required.
- Long hair, loose clothing, and jewelry must be secured back to prevent them from interfering with work.

Prohibited Items/Actions

- Absolutely no food or drink items, including chewing gum, are allowed into the lab at any time.
- Students are not permitted to bring personal items like backpacks or cell phones into the immediate lab area.
- Never use your mouth for pipetting; always use a mechanical pipetting device.

Lab Practices

- Wash your hands thoroughly with soap and water before leaving the lab.
- Disinfect your work area with the designated disinfectant (e.g., Virex) before and after each lab session.
- Aseptic technique is critical to prevent contamination. Treat all microorganisms as potential pathogens. Always flame inoculating loops/needles before and after use.
- Dispose of all contaminated materials and cultures in designated biohazard bins. Glass slides go in a specific glass disposal bin.

0.2: Lab Notebook Guidelines

All entries must be in permanent blue or black ink only. Do not use a pencil or white-out.

Table 0.2.1: Notebook Section Guidelines

Section	Definition	Expectation	Example (Gram Stain Unknown)
Purpose	A concise statement (1–3 sentences) identifying the objective.	Use professional, third-person language.	"The purpose of this experiment is to identify an unknown bacterial sample using Gram staining..."
Hypothesis	A tentative, testable prediction of the outcome.	Format as a specific "If... then..." statement.	"If the unknown is Gram-positive, then it will appear purple after staining..."
Methods	A step-by-step record of the procedure, reagents, and equipment used.	Provide enough detail for another scientist to reproduce your work. Note deviations from the manual.	1. Aseptically transferred one loopful of culture to a glass slide. 2. Heat-fixed the slide over a Bunsen burner for 3 passes. ...
Results	The raw, objective data and observations collected.	Organize data in tables/charts. Include labeled drawings for microscopic observations. Do not interpret data in this section.	Gram Stain: Purple, rod-shaped cells
Conclusion	A summary relating results back to the hypothesis and purpose.	State whether the hypothesis was supported or rejected. Discuss potential sources of error.	"The results supported the hypothesis, as the cells appeared purple (Gram-positive), suggesting the unknown is likely a Bacillus species."

CHAPTER OVERVIEW

Lab 1: Safety, Culture Handling, Microscopes, and Streak Technique

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1.1: Lab 1 Introduction

Safety, Culture Handling, Microscopes, and Streak Technique

Learning Objectives

- Understand and practice all laboratory safety procedures and the guidelines for lab notebook entries.
- Learn the proper use and focusing of a compound light microscope, including Kohler illumination setup and oil immersion technique.
- Master the aseptic technique for handling cultures and performing a streak plate for isolation of pure cultures.

Introduction

Microscopes are essential for visualizing microorganisms, while the streak plate technique is the standard method for isolating individual colonies from a mixed culture. Proper aseptic technique is crucial to prevent contamination throughout the process.

Materials

- Compound light microscopes, lens cleaner, paper, and immersion oil.
- Pre-made slides.
- Blood Agar Plates (BAP)
- Sterile cotton swabs
- Inoculating loops and incinerators

1.2: Lab 1 Protocol

Part A: Microscope Use and Kohler Illumination

1. Receive a demonstration on proper microscope operation, focusing, and cleaning.
2. Practice focusing on a pre-made slide.
3. Follow these steps to perform the Kohler illumination setup:
   1. Ensure the 10X objective lens is locked into position.
   2. Turn the brightness adjustment to its highest setting.
   3. Use the condenser knob to raise the condenser lens to its uppermost position.
   4. Close the field diaphragm entirely.
   5. Close the iris diaphragm completely.
   6. Use the coarse adjustment knob to raise the stage to its top position.
   7. Lower the stage with the coarse knob until the polygon of light comes into clear, sharp focus.
   8. If the polygon of light is not centered, use the two condenser screws to move it to the center of the field of view.
   9. Slowly open the field diaphragm until the field is completely filled with light.

Part B: Oil Immersion with Sample Slide

Follow these steps to observe stained bacterial slides using the high-magnification oil immersion lens (100X).

1. Place the slide on the stage and clip it in place.
2. Using the 10X objective, use the fine focus knob to focus the sample. (Do not move on until a clear image/color is seen!)
3. Switch to the 40X objective lens.
4. Refocus with the fine focus knob (only a small amount of adjustment should be needed).
5. Straddle the 40X and 100X lenses by rotating the nosepiece halfway between them.
6. Using the iris diaphragm, open it up until you see light coming through the physical slide (do not look through the ocular lens).
7. Place a drop of oil where you see light on the slide.
8. Move the 100X lens into the oil, ensuring the lens tip is submerged.
9. Refocus with the fine focus knob (only a small amount of adjustment should be needed).

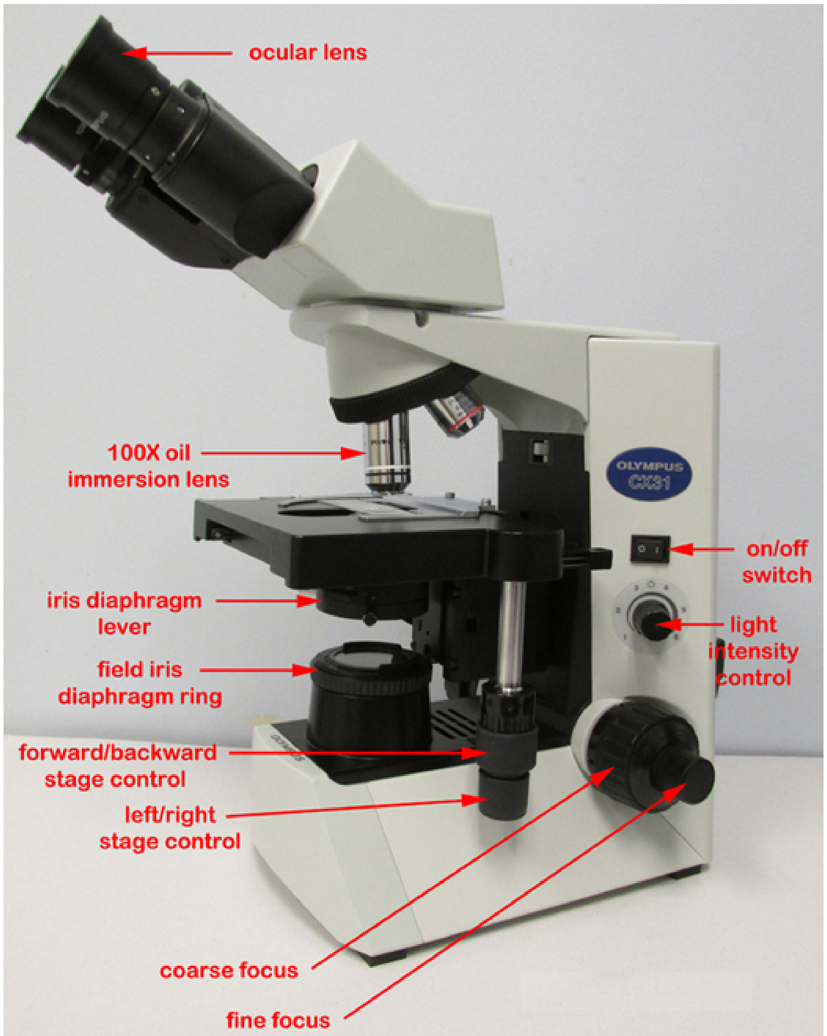


Figure 1.2.1: Diagram of a microscope with labeled parts and annotations. (Microbiology Laboratory Manual, Gary Kaiser, CC BY 4.0)

Part C: Aseptic Technique and Streak Plate

1. Observe and practice the proper technique for sterilizing the inoculating loop using the incinerator, ensuring the loop is cool before touching a culture.
2. Obtain a sterile cotton swab and collect a throat culture sample (or an alternative provided by the instructor).
3. Obtain a BAP. Label the bottom edge with your name, date, and lab section.
4. Using the swab, streak the first quadrant of the BAP.
5. Sterilize your loop and cool it.
6. Streak into the first quadrant once or twice, then streak the second quadrant.
7. Repeat the sterilization and cooling process, then streak into the third quadrant.
8. Repeat the sterilization and cooling process one final time, then streak into the fourth quadrant, being careful not to touch the first quadrant again.
9. Place labeled plates upside down in the 37°C incubator for 18-24 hours.
10. Dispose of used cotton swabs in the biohazard bag. Wipe down your lab bench with disinfectant and wash your hands before leaving the lab.

1.3: Lab 1 Notebook Entry

Complete an entry with the following information for each section:

? Notebook Entry

Purpose

Write 1-3 sentences here in third-person language, defining the objective of today's lab, including microscope setup, oil immersion, and streaking for isolation.

Hypothesis

Write an "If... then..." statement here, predicting the outcome of your streak plate technique or the clarity of your initial microscope observations.

Methods

Write a step-by-step record of the procedures used today for microscope setup, oil immersion, and preparing the streak plate. Note any deviations from the manual. You may reference the manual protocols and list only changes.

Results

Record any raw data or observations collected.

Conclusion

Write a summary that relates your results back to your hypothesis and purpose. State whether the hypothesis was supported or rejected. Discuss potential sources of error and highlight the importance of proper microscope calibration for accurate observations.

Your Notebook Entry

1.4: Microscopy Tips and Troubleshooting

Maintaining an oil immersion lens is critical for high-resolution microbiology. Below are some common mistakes, why they happen, and how to avoid or solve them.

Blurry Image on 100x

- Why it Happens: Using too little oil or trapping an air bubble.
- Avoid it/Solve It: Use only one large drop; if bubbles appear, gently rotate the nosepiece back and forth once to "spread" the oil.

Getting Oil on the 40x Lens

- Why it Happens: Switching objectives without moving the stage or rotating the wrong way.
- Avoid it/Solve It: Never go back to the 40x (dry) lens after oil has been applied to the slide; rotate the nosepiece away from the 40x to engage the 100x.

Cracked Slide or Lens

- Why it Happens: Using the coarse adjustment knob at high power.
- Avoid it/Solve It: Once you move to 40x or 100x, lock your hand to the fine adjustment knob only.

"Hazy" or Foggy View

- Why it Happens: Dried oil residue from a previous student.
- Avoid it/Solve It: Before starting, clean the 100x lens with lens paper and a drop of 70% ethanol or specialized lens cleaner.

Specimen "Disappears"

- Why it Happens: The slide is upside down.
- Avoid it/Solve It: Ensure the smear/stain is on the top side of the slide; you can focus at 10x but will never reach focus at 100x if the glass thickness is in the way.

Cleaning Quick-Reference

- Use Only: Lens Paper (soft, lint-free).
- Never Use: Kimwipes, paper towels, or shirt sleeves (these contain abrasive fibers that scratch delicate coatings).
- When cleaning at the end of class, wipe the lens in a spiral motion from the center outward.
- Always clean the 100x lens immediately after use, as dried oil becomes "gummy" and requires harsh solvents that can degrade the lens adhesive over time.

CHAPTER OVERVIEW

Lab 2: Examining Throat Cultures and Gram Staining

2.1: Lab 2 Introduction

Examining Throat Cultures and Gram Staining

Learning Objectives

- Observe and characterize the colonial morphology of microorganisms from a throat culture.
- Perform a Gram stain procedure correctly to differentiate bacteria based on cell wall composition.
- Observe and record bacterial cell shape, arrangement, and Gram reaction under oil immersion microscopy.

Introduction

The Gram stain is a fundamental differential staining technique in microbiology that separates bacteria into two major groups: Gram-positive (purple) and Gram-negative (pink) based on their cell wall properties. These results, combined with colonial and cellular morphology, provide essential information for identification.

Materials

- Incubated BAP from Safety, Culture Handling, Microscopes, and Streak Technique.
- Control agar plate with a purified known organism.
- Clean microscope slides, DI water dropper bottle, wax pencils, scum buckets
- Inoculating loops and an incinerator
- Staining rack and water source
- Bibulous paper/paper towels
- Reagents: Crystal Violet, Gram's Iodine, Decolorizer (alcohol/acetone), Safranin
- Immersion oil.

2.2: Lab 2 Protocol

Colony Morphology Observation

Observe your incubated BAP. Record characteristics like shape, margin, elevation, color, and hemolysis (clearing around colonies) for distinct colony types (at least 3) in your notebook.

Smear Preparation

1. Place a drop of water on a clean slide. Aseptically transfer a small amount of an isolated colony into the water and mix to create a thin smear.
2. Allow the smear to completely air dry.
3. Heat-fix the slide by waving it over the top of the incinerator for 20 seconds. This adheres the bacteria to the slide.

Staining Procedure

1. Place the heat-fixed slide on a staining rack over a sink.
2. Flood the smear with Crystal Violet (primary stain) for 60 seconds.
3. Rinse gently with tap water until the runoff is clear.
4. Flood the smear with Gram's Iodine (mordant) for 60 seconds. This forms a crystal violet-iodine complex within the cell wall.
5. Rinse gently with tap water until the runoff is clear.
6. Quickly decolorize with Gram's Decolorizer. Add the decolorizer drop by drop until the runoff is clear (5-10 seconds max). This is a critical step; over-decolorization can lead to false results.
7. Immediately rinse gently with water to stop the decolorization process.
8. Flood the smear with Safranin (counterstain) for 60 seconds.
9. Rinse gently with water until the runoff is clear.

Drying and Observation

Gently blot the slide dry with bibulous paper or air dry.

Microscopy

Observe the slide under the microscope, first finding the smear with 10x, then 40x. Add a drop of immersion oil and observe using the 100x oil immersion objective. Record the cell shape (cocci, rods, etc.), arrangement (diplo-, strepto-, staphylo- or singles), and color (purple for Gram-positive, pink for Gram-negative).

2.3: Lab 2 Notebook Entry

Complete an entry with the following information for each section:

Notebook Entry

Purpose

Write 1-3 sentences here in third-person language, defining the objective of today's lab..

Hypothesis

Write an "If... then..." statement here, predicting the outcome of your Gram stain or culture observations based on existing knowledge.

Methods

Write a step-by-step record of the procedures used today. Note any deviations from the manual. You may reference the manual protocols and list only changes.

Results

Record all raw data and observations here. Use tables for colonial morphology and staining results. Include labeled drawings of what you see under the 100x microscope objective. Do not interpret data in this section.

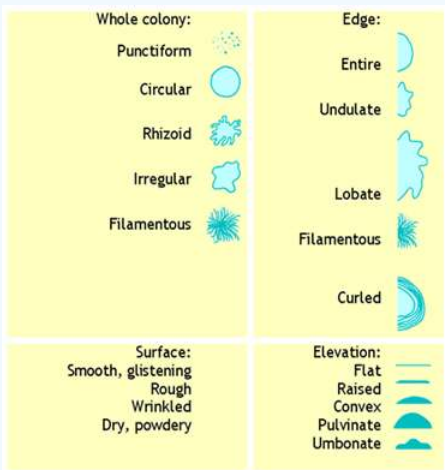
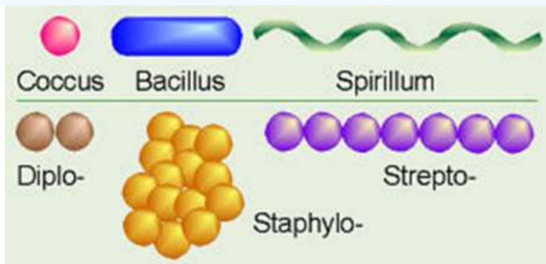


Figure 2.3.1: Colony shapes. (Wikimedia)



igure 2.3.2: Bacterial shapes. (Wikimedia)

Colony Observation Description

- Shape (e.g., Circular, irregular)
- Margin (e.g., Entire, undulate)
- Elevation (e.g., Raised, flat)
- Color/Hemolysis (e.g., White, Alpha/Beta/Gamma)

Microscopic Observation Description

- Gram Reaction (e.g., Gram-positive (purple), Gram-negative (pink)
- Cell Shape (e.g., Coccus, Bacillus)
- Arrangement (e.g., Diplo-, Staphylo-, Strepto-)

Conclusion

Write a summary that relates your results back to your hypothesis and purpose. State whether the hypothesis was supported or rejected. Discuss potential sources of error or next steps for identification.

Your Notebook Entry

CHAPTER OVERVIEW

Lab 3: Tiny Earth Introduction and Planning

3.1: Lab 3 Introduction

Tiny Earth Introduction and Planning

Learning Objectives

- Understand the global health crisis of antibiotic resistance and the purpose of the Tiny Earth project.
- Learn the importance of soil as a source of antibiotic-producing bacteria.
- Begin planning for soil collection and experimental design.

Introduction

Antibiotic resistance is a serious global problem, with few new drugs being developed by pharmaceutical companies. The Tiny Earth project aims to crowd-source the discovery of new antibiotics from soil microbes, as many current antibiotics originally came from soil bacteria. Students will engage in real research by collecting soil, isolating bacteria, and testing for antibiotic production.

Materials

- Soil Collection Kit (Ziploc bag, 50mL conical tube or whirl bag, paper ruler, scoopula or spoon)
- Calculators, Sharpies (in room)

3.2: Lab 3 Protocol

Planning Discussion

Discuss in groups or individually about potential soil collection locations and the type of environment that might yield diverse bacteria.

Hypothesis Development

Formulate hypotheses about the abundance and diversity of microbes in your chosen soil sample.

Methodological Design

Devise a method to transfer microbes from the soil sample to a lab medium. Consider the need for dilution methods to obtain single colonies.

Data Recording

Note down your preliminary plan and hypothesis in your notebook.

3.3: Lab 3 Notebook Entry

Complete an entry with the following information for each section:

Notebook Entry

Purpose

Write 1-3 sentences here in third-person language, defining the objective of today's lab.

Hypothesis

Write an "If... then..." statement here, predicting which type of environment will provide the most diverse or antibiotic-producing bacteria.

Methods

Write a step-by-step description of the planned method for collecting soil and transferring it to lab media (e.g., using serial dilutions to obtain single colonies). Note any specific materials needed.

Results

Record any initial observations or calculations from your planning phase here. You may list the items in your soil collection kit.

Conclusion

Write a summary of your planning session. Discuss how your plan addresses the hypothesis and potential challenges in isolating single colonies from soil.

Your Notebook Entry

CHAPTER OVERVIEW

Lab 4: Soil Collection Field Work

4.1: Lab 4 Introduction

Soil Collection Field Work

Learning Objectives

- Collect a soil sample from a chosen location using a sterile technique.
- Record detailed environmental and location data for the sample.

Introduction

Collecting soil from a specific, well-documented environment is crucial for tracking the origin of potential antibiotic producers. Students will use the provided kit and smartphone to collect their sample and record important environmental information.

Materials

- Soil collection kit
- Smartphone (for GPS coordinates and photo)
- Paper/Writing utensil

4.2: Lab 4 Protocol

Plan Location

Finalize and travel to your planned soil collection location.

Collection

Using materials in the kit, collect about a handful of soil in the sterile tube or bag.

GPS Coordinates

Use a smartphone to obtain the GPS coordinates of the location.

Fill Out the Data

Fill out the soil collection table in your notebook with all required information (General Location, GPS Coordinates, Date & Time, Sample Site Descriptors, Air Temp (°C), Depth (in), Type of Soil). Use a soil identification chart if available.

Develop Hypotheses

Refine hypotheses about how the soil will be used and what can be discovered.

4.3: Lab 4 Notebook Entry

Complete an entry with the following information for each section:

Notebook Entry

Purpose

Write 1-3 sentences here in third-person language, defining the objective of today's lab.

Hypothesis

Write an "If... then..." statement here, predicting the abundance and diversity of microbes in your specific soil sample based on the site conditions.

Methods

Write a step-by-step record of the procedures used today for soil collection and data entry. Note any deviations from the manual. You may reference the manual and list only changes.

Results

Record all raw data and observations here. Include the completed soil collection table with all site descriptors and environmental data.

Conclusion

Write a summary that relates your results back to your hypothesis and purpose. State whether the hypothesis was supported or rejected. Discuss potential sources of error during collection and how this data will be used in future labs.

Your Notebook Entry

CHAPTER OVERVIEW

Lab 5: Plating Soil Samples

5.1: Lab 5 Introduction

Plating Soil Samples

Learning Objectives

- Perform serial dilutions of a soil sample to reduce bacterial concentration.
- Plate diluted soil samples to obtain single colonies for isolation.

Introduction

Because a single gram of soil can contain billions of bacterial cells, serial dilutions are essential to obtain isolated, pure colonies for study. Students will perform dilutions and spread the samples on agar plates to encourage single colony formation.

Materials

- Soil samples from Soil Collection Field Work
- Glass test tubes pre-marked for 9mLs
- 1.5mL microcentrifuge tubes
- 1x PBS for dilutions
- 100-1000ul pipettors with tips
- Disposable plastic pipettes and L-shaped spreaders
- LB + antifungal plates (up to 4 per student)
- Scale/Balance

5.2: Lab 5 Protocol

Serial Dilution

1. Weigh out 1g of your soil sample and add it to 9mL of 1x PBS to make a 1:10 dilution.
2. Transfer 100ul from the 1:10 dilution to 900ul of 1x PBS in a microcentrifuge tube to make a 1:100 dilution.
3. Repeat this process to create further dilutions (e.g., 1:1000, 1:10000).

Plating

1. Pipette 100ul from your last few dilutions onto separate LB plates (each student plates 3-4 dilutions).
2. Use a sterile spreader to spread the dilution over the entire plate. Go from the most diluted to the least diluted to use one spreader.

Incubation

Incubate plates at room temperature.

Cleanup

Dispose of tips and spreaders in the biohazard. Return metal scoopulas.

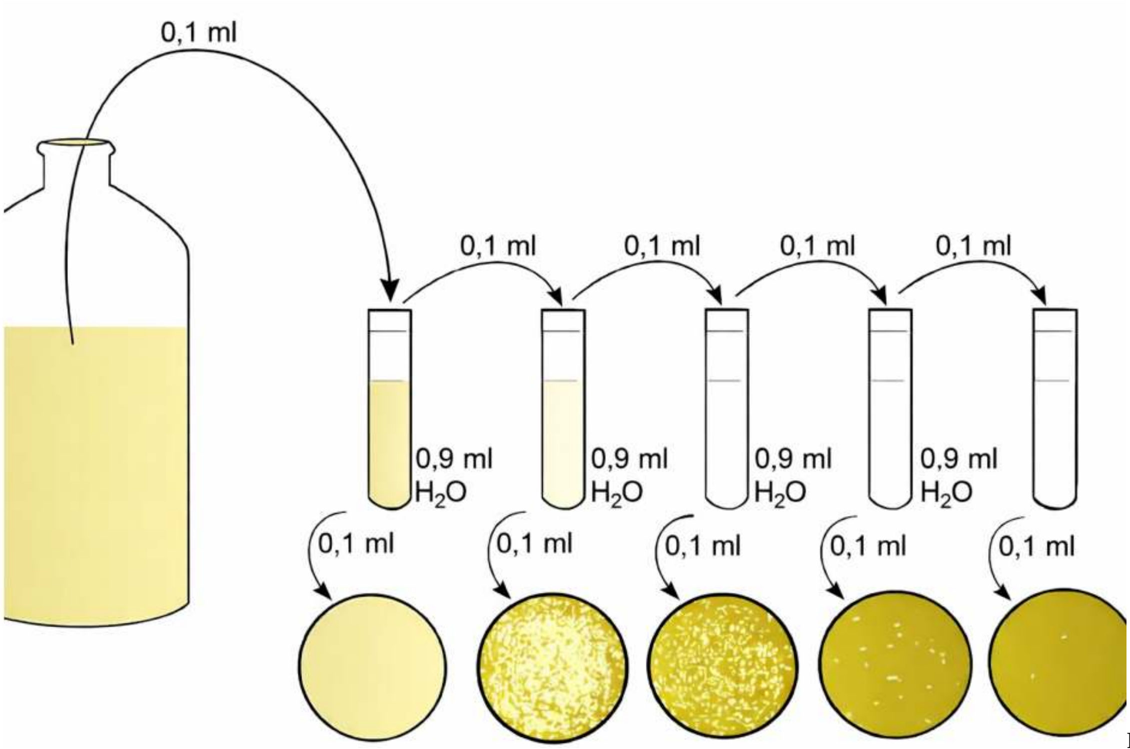


Figure 5.2.1: Diagram of a serial dilution process. (Wikimedia).

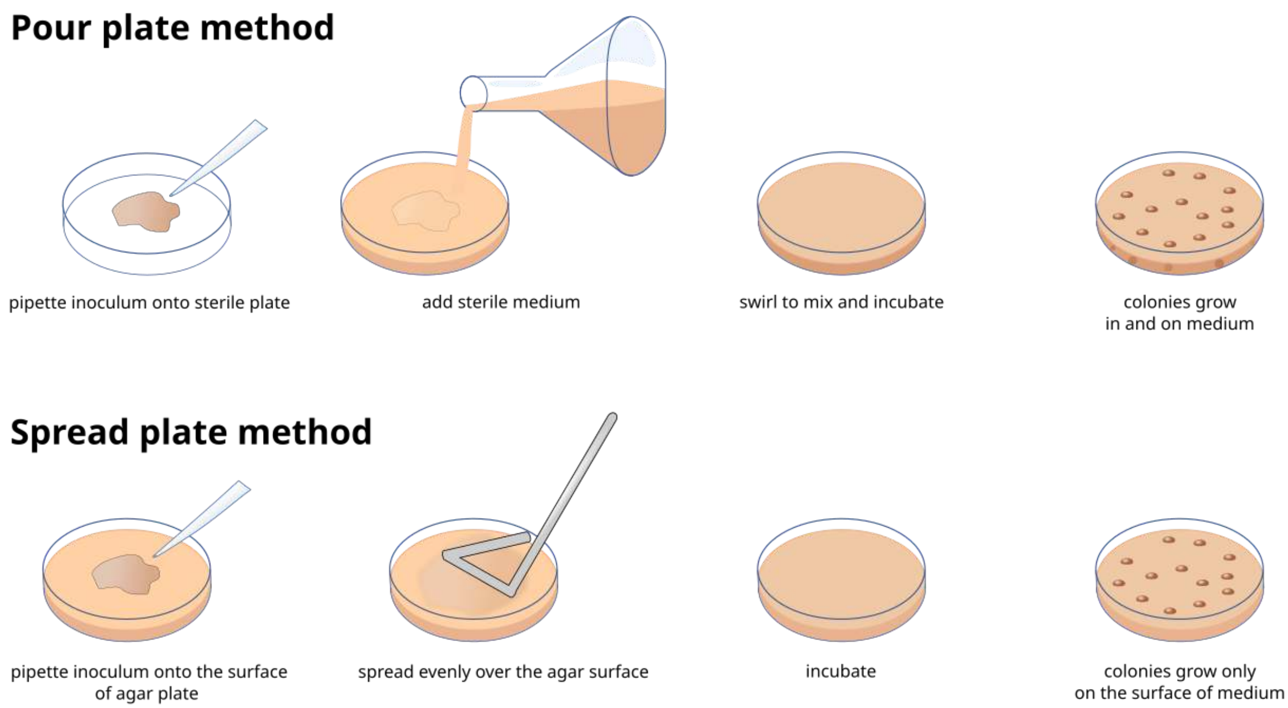


Figure 5.2.2: Illustration of the Spread Plate Method. (Wikimedia).

5.3: Lab 5 Notebook Entry

Complete an entry with the following information for each section:

? Notebook Entry

Purpose

Write 1-3 sentences here in third-person language, defining the objective of today's lab.

Hypothesis

Write an "If... then..." statement here, predicting which dilution will yield the optimal number of isolated colonies (50-100 colonies).

Methods

Write a step-by-step record of the procedures used today. Note any deviations from the manual. You may reference the manual and list only changes.

Results

Record all raw data and observations here. After incubation, count the colonies on your plates and record the counts. Note the final dilution factor for each plate (e.g., 1:1000 final dilution if 100uL of 1:100 dilution was plated).

Conclusion

Write a summary that relates your results back to your hypothesis and purpose. State whether the hypothesis was supported or rejected. Discuss potential sources of error and determine which dilution concentration is best for moving forward.

Your Notebook Entry

CHAPTER OVERVIEW

Lab 6: Bacterial Culture Conditions

6.1: Lab 6 Introduction

Bacterial Culture Conditions

Learning Objectives

- Understand that bacteria have specific environmental condition preferences for growth.
- Differentiate between various forms of culture media (solid, semi-solid, liquid) and their applications.
- Select appropriate media and environmental conditions to encourage the growth of diverse bacteria from soil samples.

Introduction

Bacteria can thrive in diverse and extreme conditions, with each species having a specific preference for temperature, pH, light, and oxygen availability. To study bacteria in a lab, appropriate media must be used, which can be general-purpose, enriched, selective, or differential. This lab allows students to explore these concepts by choosing different culture conditions for their soil bacteria and observing the effects on growth.

Materials

- Student dilutions from Plating Soil Samples
- Multiple types of media with antifungal: PDA, TSA, R2A, LB agar
- Pipettor and tips
- Disposable L-shaped spreaders

6.2: Lab 6 Protocol

Calculate CFU

Determine the Colony Forming Units (CFU) for your plated dilutions from the last lab using the formula: Number of colonies x dilution factor inverse = CFU/g.

Select Dilution and Conditions

1. Choose the soil dilution concentration that yielded between 50 and 100 colonies.
2. Select two different culture conditions from the available options (e.g., a specific type of agar or an environmental change like temperature or darkness).

Table 6.2.1 Culture Conditions

Agar Type	Key Ingredients (per Liter)	Primary Use
PDA (Potato Dextrose Agar)	Potato infusion (from 200g potatoes), Dextrose (20g), Agar (15g)	General-purpose medium for the **cultivation and enumeration of yeasts and molds**. Its low pH (5.6) naturally inhibits many bacteria, and it can be further acidified for selectivity.
TSA (Tryptic Soy Agar)	Pancreatic digest of casein (15g), Peptic digest of soybean meal (5g), NaCl (5g), Agar (15g)	**General-purpose, non-selective medium** for the isolation and cultivation of a wide variety of both fastidious and non-fastidious bacteria, as well as many yeasts and molds. Used for sterility testing and microbial monitoring.
R2A (Reasoner's 2A Agar)	Yeast extract, proteose peptone, casein hydrolysate, dextrose, soluble starch, sodium pyruvate, various salts, Agar (low nutrient levels overall)	**Enumeration and cultivation of bacteria from potable water**. Its low nutrient content, combined with longer and cooler incubation times, is ideal for recovering stressed, slow-growing bacteria often found in treated water supplies.
LB (Luria-Bertani) Agar	Tryptone (10g), Yeast extract (5g), NaCl (5-10g), Agar (15g)	A rich medium widely used in **molecular biology and genetic research** for the general culture, maintenance, and propagation of *Escherichia coli* and other members of the Enterobacteriaceae. Antibiotics are often added to select for specific genetic modifications.

Plate Samples

1. Pipette approximately 100ul of the selected dilution onto each of the two new plates.
2. Use a sterile spreader to spread the dilution over the entire plate.

Incubate Plates

Incubate the plates at the selected temperature or condition.

Cleanup

Dispose of tips and spreaders in the biohazard container.

6.3: Lab 6 Notebook Entry

Complete an entry with the following information for each section:

Notebook Entry

Purpose

Write 1-3 sentences here in third-person language, defining the objective of today's lab. Hint: Focus on how environmental conditions affect bacterial growth and identification.

Hypothesis

Write an "If... then..." statement here, predicting how your chosen culture conditions will affect the growth and morphology of your soil bacteria compared to the previous lab's results.

Methods

Write a step-by-step record of the procedures used today, including the CFU calculation and plating process. Note any deviations from the manual. You may reference the manual and list only changes.

Results

Record all raw data and observations here. Document your CFU calculations. After incubation, describe and record the characteristics of colonies on your plates, noting color, texture, edge, and elevation. Compare these observations to those from Plating Soil Samples.)

Conclusion

Write a summary that relates your results back to your hypothesis and purpose. State whether the hypothesis was supported or rejected. Discuss potential sources of error and summarize the optimal conditions for growing the bacteria you isolated.

Your Notebook Entry

CHAPTER OVERVIEW

Lab 7: The Master Plate and Antibiotic Screening

7.1: Lab 7 Introduction

Safety, Culture Handling, Microscopes, and Streak Technique

Learning Objectives

- Isolate unique bacterial colonies from a mixed environmental sample using the pick and patch method.
- Create a "master plate" to organize isolates for further testing.
- Screen isolates for antibiotic production against known tester strains using the spread-patch method.

Introduction

- Isolate unique bacterial colonies from a mixed environmental sample using the pick and patch method.
- Create a "master plate" to organize isolates for further testing.
- Screen isolates for antibiotic production against known tester strains using the spread-patch method.

Materials

- Your incubated dilution plates from Plating Soil Samples and Bacterial Culture Conditions.
- Sterile toothpicks
- LB agar plates w/ antifungal (1 per student for master plate)
- LB plates (2 per student for screening)
- BHI broth cultures of *Bacillus subtillis* and *Erwinia carotovora* (tester strains)
- *Lysobacter antibioticus* culture (positive control)
- Disposable L-shaped spreaders
- Pipettors and tips

7.2: Lab 7 Protocol

Part A: Making the Master Plate

1. Examine your dilution plates from the previous lab, noting how conditions affected growth.
2. Draw a 20-box grid on the back of an LB agar plate. Label as "Master Plate" and number the boxes 1-20.
3. Using a sterile toothpick, gently touch a single, isolated colony from your dilution plates. Avoid known bacteria, as pointed out by your instructor.
4. Transfer the bacteria to a box on the master plate by making a squiggle or "X". Record the colonial phenotype (color, texture, edge, elevation) in your notebook for this location.
5. Repeat using a fresh toothpick for each unique colony until all boxes are filled (or you run out of unique colonies). Dispose of toothpicks in the biohazard.
6. Incubate the master plate at room temperature.

Part B: Screening for Antibiotic Production

1. Obtain two fresh LB plates. Draw the same grid on the back of both.
2. Label one plate "G+ Screen" (for *B. subtilis*) and the other "G- Screen" (for *E. carotovora*).
3. Pipette 100ul of the G+ tester strain broth onto its plate and spread evenly across the entire surface to form a "lawn". Let dry.
4. Repeat for the G- tester strain on its plate.
5. Using the same pick and patch method as above, transfer your isolates from the master plate to the corresponding boxes on both G+ and G- plates. Ensure the orientation matches the master plate layout.
6. Include a patch of the positive control bacterium (*L. antibioticus*) on each plate if space allows.
7. Incubate plates at room temperature.

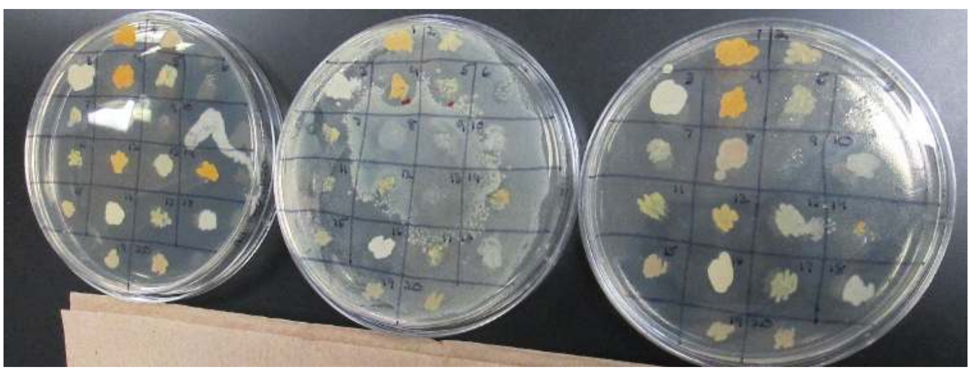


Figure 7.2.1: Petri dishes with bacterial colonies. (Photo courtesy of Angelo Kolokithas, 2025)

7.3: Lab 7 Notebook Entry

Complete an entry with the following information for each section:

? Notebook Entry

Purpose

Write 1-3 sentences here in third-person language, defining the objective of today's lab.

Hypothesis

Write an "If... then..." statement here, predicting which isolates might show antibiotic production and why, or generally how the screening process works.

Methods

Write a step-by-step record of the procedures used today for both making the master plate and screening the isolates. Note any deviations from the manual. You may reference the manual and list only changes.

Results

Record all raw data and observations here. Create a table of your master plate layout, listing the phenotype (color, texture, edge, etc.) for each isolate number. After incubation of the screening plates, use another table or a diagram of the plates to record the presence and size of any zone of inhibition observed for each isolate against both G+ and G- tester strains.

Conclusion

Write a summary that relates your results back to your hypothesis and purpose. State whether the hypothesis was supported or rejected. Discuss potential sources of error and highlight which isolates were identified as potential antibiotic producers.

Your Notebook Entry

CHAPTER OVERVIEW

Lab 8: Broad-Spectrum Screening

8.1: Lab 8 Introduction

Broad-Spectrum Screening

Learning Objectives

- Identify promising antibiotic-producing isolates from initial screening results.
- Test selected producers against a panel of 9 ESKAPE pathogen relatives to determine the spectrum of activity.
- Generate a fresh, pure culture of promising isolates for downstream applications (PCR).

Introduction

Once an isolate shows initial antibiotic activity, it must be tested more broadly to determine its effectiveness. By testing against multiple pathogens, we can classify the antibiotic as broad-spectrum (effective against many bacteria) or narrow-spectrum (effective against a few). The size of the zone of inhibition correlates to the sensitivity of the tester strain to the isolate's antibiotic.

Materials

- Your screening plates and master plates from The Master Plate & Antibiotic Screening.
- Liquid cultures of all 9 ESKAPE pathogen relatives (e.g., *S. epidermidis*, *E. carotovora*, etc.)
- LB plates w/ antifungal (2 per student)
- Inoculating loops, needles, and an incinerator
- Pipettors and tips
- Sterile toothpicks

8.2: Lab 8 Protocol

Review Results

Examine your G+ and G- screening plates from Lab 7. Identify the isolate(s) that produced the largest zone of inhibition or showed inhibition against both tester strains. Note any colonies on the master plate that changed morphology over time.

Pinwheel Plate Setup

1. Obtain one fresh LB plate. Draw a "pinwheel" with 9 equal sections on the back of the plate using a marker, like spokes on a wheel. Label each section for one of the 9 ESKAPE pathogens provided by the instructor.
2. Using a sterile loop, inoculate a "lawn" of each specified pathogen within its corresponding section, taking care to keep within the lines.
3. Allow the plates to dry.

Cross-Streaking

Using a sterile toothpick, transfer a chunk of your chosen antibiotic-producing colony to the very center of the plate, touching the edge of the lawn of each of the 9 sections.

Incubation

Incubate the pinwheel plate at room temperature.

Pure Culture Preparation

Obtain a second fresh LB plate. Use a sterile loop to perform a streak plate technique (from the Safety, Culture Handling, Microscopes, and Streak Techniques lab) with your chosen antibiotic producer to obtain a fresh, pure streak plate. This culture will be used for PCR in the next lab.

Incubation (Pure Culture)

Incubate the new streak plate at room temperature.

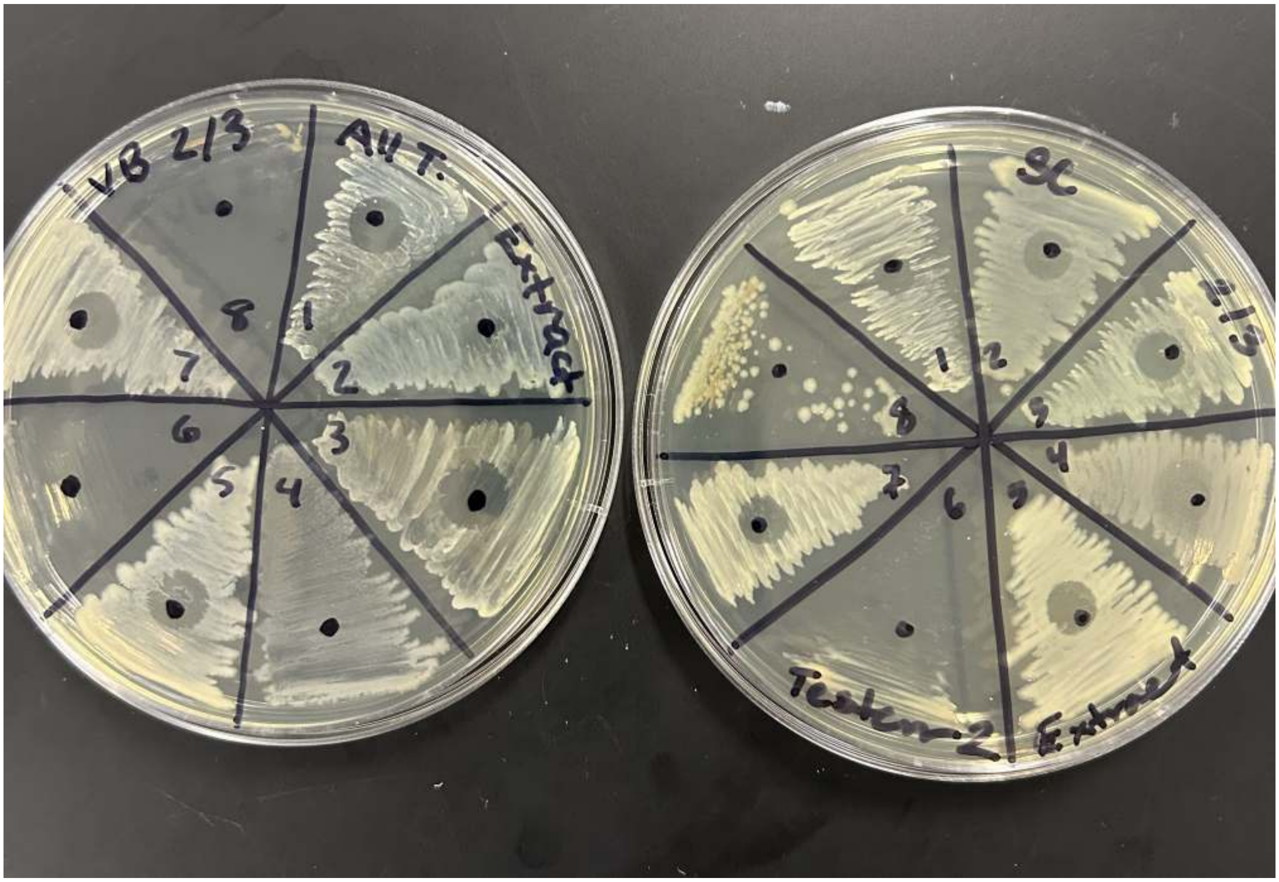


Figure 8.2.1: Petri dish with bacterial colonies showing zones of inhibition. (Photo courtesy of Angelo Kolokithas)

8.3: Lab 8 Notebook Entry

Complete an entry with the following information for each section:

Notebook Entry

Purpose

Write 1-3 sentences here in third-person language, defining the objective of today's lab.

Hypothesis

Write an "If... then..." statement here, predicting which of the 9 pathogens will be sensitive to your chosen isolate.

Methods

Write a step-by-step record of the procedures used today for setting up the pinwheel plate and the fresh streak plate. Note any deviations from the manual. You may reference previous manual protocols and list only changes.

Results

Record all raw data and observations here. Create a diagram of your pinwheel plate and, after incubation, record the presence and size (in mm) of any zone of inhibition for each of the 9 pathogens tested. Note the growth of your fresh streak plate.

Conclusion

Write a summary that relates your results back to your hypothesis and purpose. State whether the hypothesis was supported or rejected. Discuss potential sources of error and characterize the broad- or narrow-spectrum activity of your isolate.

Your Notebook Entry

CHAPTER OVERVIEW

Lab 9: Colony PCR

9.1: Lab 9 Introduction

Colony PCR

Learning Objectives

- Understand the principles of the Polymerase Chain Reaction (PCR) and DNA amplification.
- Perform a colony PCR to amplify the 16S ribosomal DNA gene from an isolated bacterium.
- Prepare PCR samples for gel electrophoresis analysis.

Introduction

Polymerase Chain Reaction (PCR) is a technique used to amplify specific sections of DNA. This process uses an enzyme (DNA polymerase), primers (short DNA sequences that target the 16S rRNA gene, which is highly conserved across all bacteria), and a thermal cycler to rapidly create millions of copies of the target DNA sequence. The 16S gene sequence is used for bacterial identification.

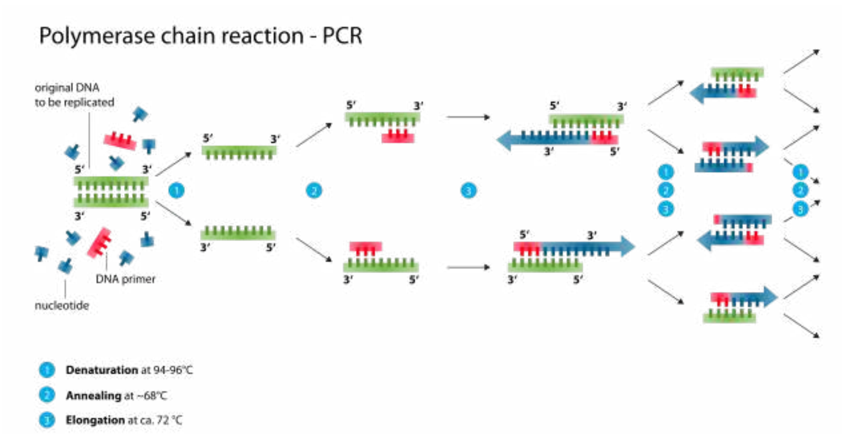


Figure 9.1.1: Polymerase Chain Reaction. (Wikimedia).

Materials

- Your fresh streak plate from Broad-Spectrum Screening.
- 0.2 mL dome-topped PCR tube with 25ul of PCR Master Mix (contains polymerase, primers, buffer, dyes)
- Pipettors and tips

9.2: Lab 9 Protocol

Labeling Tubes

Obtain a 0.2 mL PCR tube. Label only the dome/cap of the tube with your initials. Writing on the side will rub off.

Adding Bacteria

1. Using a sterile P200 pipettor tip or a sterile stab at a 25° angle, gently touch the tip into your desired colony. You only need a tiny, visible amount ("less is more").
2. Swirl the tip into the PCR mix to thoroughly mix the cells into the solution until there are no visible chunks.
3. Gently flick the tube to remove any bubbles.

Thermal Cycling

Give your labeled tube(s) to the instructor to be placed into the thermal cycler machine for the PCR run. Ensure your name and lab section are noted on the log sheet for storage.

Storage

After the run is complete, the instructor will store the tubes at 4°C in the refrigerator until the next lab.

9.3: Lab 9 Notebook Entry

Complete an entry with the following information for each section:

Notebook Entry

Purpose

Write 1-3 sentences here in third-person language, defining the objective of today's lab.

Hypothesis

Write an "If... then..." statement here, predicting whether your PCR reaction will successfully amplify DNA.

Methods

Write a step-by-step record of the procedures used today for preparing your PCR tube(s). Note any deviations from the manual. You may reference the manual and list only changes.

Results

Record all raw data and observations here. Note the tube label, the source of the bacteria, and predict what the result might look like when the gel is run next lab (e.g., will a band be present?).

Conclusion

Write a summary that relates your results back to your hypothesis and purpose. State whether the hypothesis was supported or rejected. Discuss potential sources of error and the theoretical principles of PCR.

Your Notebook Entry

CHAPTER OVERVIEW

Lab 10: Gel Electrophoresis and Sequencing Prep

10.1: Lab 10 Introduction

Gel Electrophoresis and Sequencing Prep

Learning Objectives

- Separate PCR products by size using agarose gel electrophoresis.
- Visually confirm successful DNA amplification using a Gel Doc system.
- Understand the principles of and prepare samples for DNA sequencing.

Introduction

Agarose gel electrophoresis uses an electric current to move negatively charged DNA fragments through a gel matrix. Smaller fragments move faster than larger ones, allowing separation by size. We use a DNA stain to visualize the bands under blue light. Successful PCR should yield a single band around 1100-1200 base pairs (bp). Only samples with a visible band will be "cleaned up" enzymatically to remove leftover primers and nucleotides before sequencing.

Materials

- Your PCR samples from Colony PCR.
- 1% agarose gel with Midori Green stain (pre-cast)
- Gel electrophoresis apparatus, 1x TAE buffer.
- Standard Ladder (DNA ladder of known sizes)
- Pipettors and tips
- BlueView Transilluminator.

10.2: Lab 10 Protocol

Part A: Running the Gel

1. Place the pre-cast gel (in its plastic mold) into the electrophoresis machine with lanes toward the top (negative electrode).
2. Pour 1x TAE buffer over the gel until it is just submerged.
3. Load 10 ul of the 1kb Standard Ladder into the first lane.
4. Load 10ul of your PCR sample into the next available lane. Keep a record of your lane number.
5. Once all samples are loaded, place the lid on the machine and ensure the power switch is set to run DNA from negative to positive (top to bottom). Run the gel at 100V for 25-40 minutes.
6. View the gel under blue light using the BlueView Transilluminator and record your results (visible bands, size, etc.).
7. Dispose of the gel in the biohazard bag.

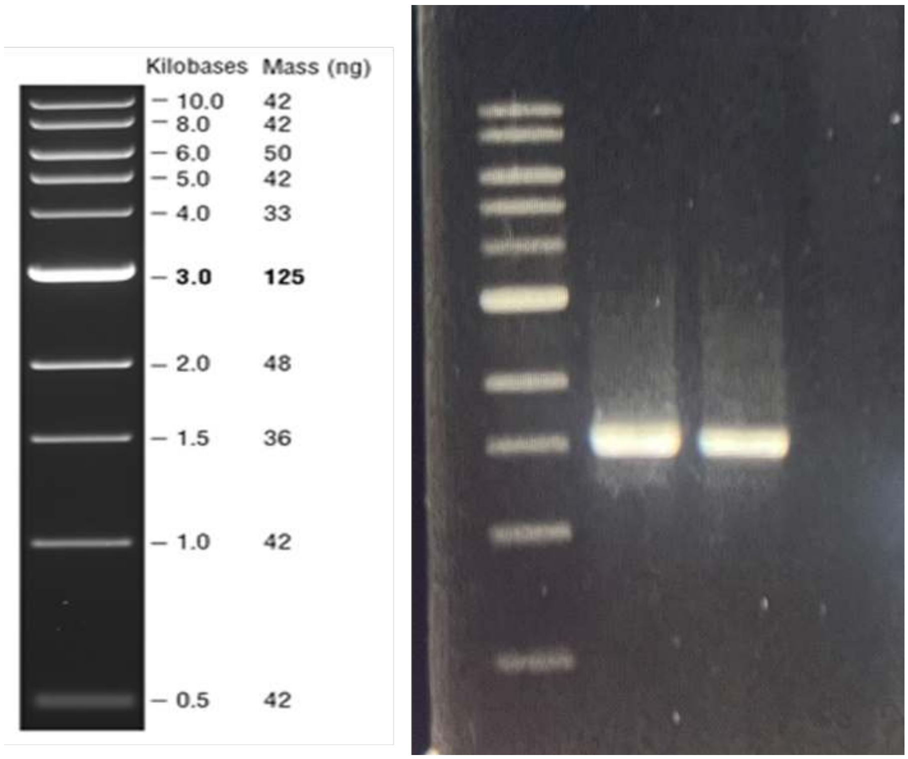


Figure 10.2.1: Sample Gel with Primers 27F and 1492R. (Photo courtesy of Drew Laux).

Part B: Sequencing Prep (Only if Part A was successful)

1. Place your sample (in order) into the "Ready for Sequencing" rack provided.
2. If successful, fill out the Sequencing form with Initials+ Colony number, class day, start time, semester, and year, and then write the corresponding spot number on the side of the tube

Part C: BLAST results (Only if Part B was successful)

1. Your instructor will notify you when the sequencing reactions data are in and ready for your review via a Canvas announcement. Find the attached file, download the file, locate your sequence, and have a BLAST by going to the BLAST site, clicking on Nucleotide blast, and pasting your sequence into the search box. After that, hit the BLAST button. (NOTE: Open the file, as the preview does not contain the entire sequence)
2. Once that is complete, it will compare your code to the genetic code of all known bacteria on the planet (takes about a minute)!
3. It will then display what it thinks is the best match, much like a Google search. You want to record the scientific name, accession, Per. Ident, and the Acc. Len columns, as well as the description. For best results, use the first full name you find with both the genus and species. For instance, if your search result's scientific name was Pseudomonas sp. (which stands for species), keep going down the list until it matches with a species, like Pseudomonas putida.

4. This information tells you what your bacteria is most closely related to (or what it is if it's a 100% match). With this information, you can further research what it is and how it compares to your phenotypic results (Gram stain, shape, grouping, etc.). You can also determine (the fun part) whether it has ever been shown to produce antibiotics by researching the name of your bacteria.

10.3: Lab 10 Notebook Entry

Complete an entry with the following information for each section:

? Notebook Entry

Purpose

Write 1-3 sentences here, defining the objectives of separating DNA by size and preparing it for sequencing.

Hypothesis

Write an "If... then..." statement here, predicting whether your gel will show a band and its approximate size, and whether you will proceed to the purification step.

Methods

Write a step-by-step record of the procedures used today for running the gel and (if applicable). Note any deviations from the manual. You may reference the manual and list only changes.

Results

Record all raw data and observations here. Draw a diagram of the gel showing the ladder and your sample lane(s). Note the presence, absence, and size (in bp) of any bands. Note if you proceeded to the purification step, and which tube number was used for sequencing.

Conclusion

Write a summary that relates your results back to your hypothesis and purpose. State whether the hypothesis was supported or rejected. Discuss potential sources of error and whether your DNA was successfully amplified and sent for sequencing. Include the identification of your microbe if sequencing was successful.

Your Notebook Entry

CHAPTER OVERVIEW

Lab 11: Glycerol Stocks

11.1: Lab 11 Introduction

Glycerol Stocks

Important

This is lab is only for successful isolates. If not successful, you can skip this entry.

Learning Objectives

- Understand the purpose of long-term bacterial preservation.
- Prepare a glycerol stock of your antibiotic-producing isolate for storage at -80°C.
- Log your sample into the lab inventory system.

Introduction

Freezing bacteria without protection causes the formation of ice crystals that rupture the cell membrane. Glycerol acts as a cryoprotectant, preventing ice formation and allowing bacteria to be stored at -80°C for years. This lab ensures that successful antibiotic producers can be preserved for future research.

Materials

- Your pure antibiotic-producing isolate on LB agar
- 20-25% Glycerol in LB Broth solution
- 2 mL Cryotubes
- Sterile toothpicks
- Pipettors and tips
- Cryoboxes and inventory log sheet

11.2: Lab 11 Protocol

Labeling Cryotube

Label a 2 mL cryotube clearly on the side and cap with your initials + colony number, lab day + start time, and semester/year (e.g., DL 12 Tues 1130 Fall 2026).

Adding Glycerol Broth

Pipette 1000ul (1 mL) of the glycerol/LB broth solution into your labeled cryotube.

Adding Bacteria

Using a sterile toothpick, pick up an isolated colony from your pure plate and aseptically mix it into the glycerol/LB broth in your tube. Swirl the toothpick to remove as much bacteria as possible.

Mixing

Gently pipette the mixture up and down or vortex the tube until the bacteria are thoroughly mixed throughout the broth.

Storage & Logging

Place your tube into the designated cryobox, in order by the number/position assigned on the inventory log sheet provided by the instructor. Fill out the log sheet with all details.

Place the cryobox in the refrigerator; lab techs will transfer it to the -80°C freezer.

11.3: Lab 11 Notebook Entry

Complete an entry with the following information for each section:

❓ Notebook Entry

Purpose

Write 1-3 sentences here, defining the objective of preserving your bacterial isolate for long-term storage.

Hypothesis

Write an "If... then..." statement here, predicting the outcome of the preservation process.

Methods

Write a step-by-step record of the procedures used today for preparing the glycerol stock. Note any deviations from the manual. You may reference the manual and list only changes.

Results

Record all raw data and observations here. Note the cryotube label information and the position number recorded on the inventory log sheet.

Conclusion

Write a summary that relates your results back to your hypothesis and purpose. State whether the hypothesis was supported or rejected. Discuss potential sources of error and the importance of this preservation method for future research.

Your Notebook Entry

CHAPTER OVERVIEW

Lab 12: Classic vs Modern Biochemical Tests

12.1: Lab 12 Introduction

Classic vs Modern Biochemical Tests

Learning Objectives

- Perform and interpret a range of classical biochemical tests for bacterial identification.
- Differentiate bacteria based on enzymatic activities and metabolic pathways (e.g., fermentation, respiration, motility, starch hydrolysis).
- Use test results to begin the process of identifying your unknown isolate.

Introduction

Before modern sequencing, microbiologists used a series of biochemical tests to identify unknowns. Results from these tests (e.g., Catalase, Oxidase, Hemolysis, Starch Hydrolysis, TSI, Motility) provide clues about a bacterium's metabolism and cell properties. We will use these tests to identify our unknown when sequencing results are pending or if sequencing was unsuccessful.

Materials

- Your pure unknown culture isolate.
- Known positive and negative control cultures/examples.
- Glass slides, H_2O_2, oxidase reagent, starch agar, iodine solution, BAP, TSI slants, motility agar tubes, OF dextrose, mineral oil,
- Loops, stabs, incinerator, microscope supplies

12.2: Lab 12 Protocol

Follow the abbreviated protocols provided below for the tests required for your isolate (Gram-positive or Gram-negative path).

Table 12.2.1: Isolate Protocols

Test	Protocol Summary	Positive Result	Negative Result
Catalase	Mix the colony with 3% H2O2 on the slide.	Rapid bubbles	No bubbles
Hemolysis	Streak on BAP, incubate, and observe clearing.	Clear zone (β); green hue (α)	No change (γ)
Starch Hydrolysis	Streak on starch agar, incubate, and add Gram's iodine.	Colorless zone around colonies	No colorless zone around colonies
TSI Slant	Stab butt, streak slant surface, incubate.	Yellow butt/slant (A/A); black butt (H2S); gas	Red butt/slant (K/K); red slant/yellow butt (K/A)
SIM	Stab straight into deep agar, incubate.	Motility: Turbidity/cloudy agar H2S: Black color Indole: Red after addition of 5 drops of Kovac's reagent	Motility: Growth only along the stab line H2S: no color change Indole: no color after addition of 5 drops of Kovac's reagent
Oxidase	Mix colony onto the oxidase reagent swab.	Blue color formation	No color change
OF dextrose	Stab straight into deep agar, incubate (x2). Overlay 1 tube with 20 drops mineral oil	Yellow color change Oxygen tube: Yellow, Fermentation tube Green=Oxidizer Oxygen tube: Yellow, Fermentation tube Yellow=fermenter	No color change

12.3: Lab 12 Notebook Entry

Complete an entry with the following information for each section:

Notebook Entry

Purpose

Write 1-3 sentences here, defining the objective of running various biochemical tests on your unknown isolate.

Hypothesis

Write an "If... then..." statement here, predicting the outcomes of these tests for your unknown organism.

Methods

Write a step-by-step record of the procedures used today across all tests. Note any deviations from the manual. You may reference the abbreviated protocol table above and list only specific changes or details.

Results

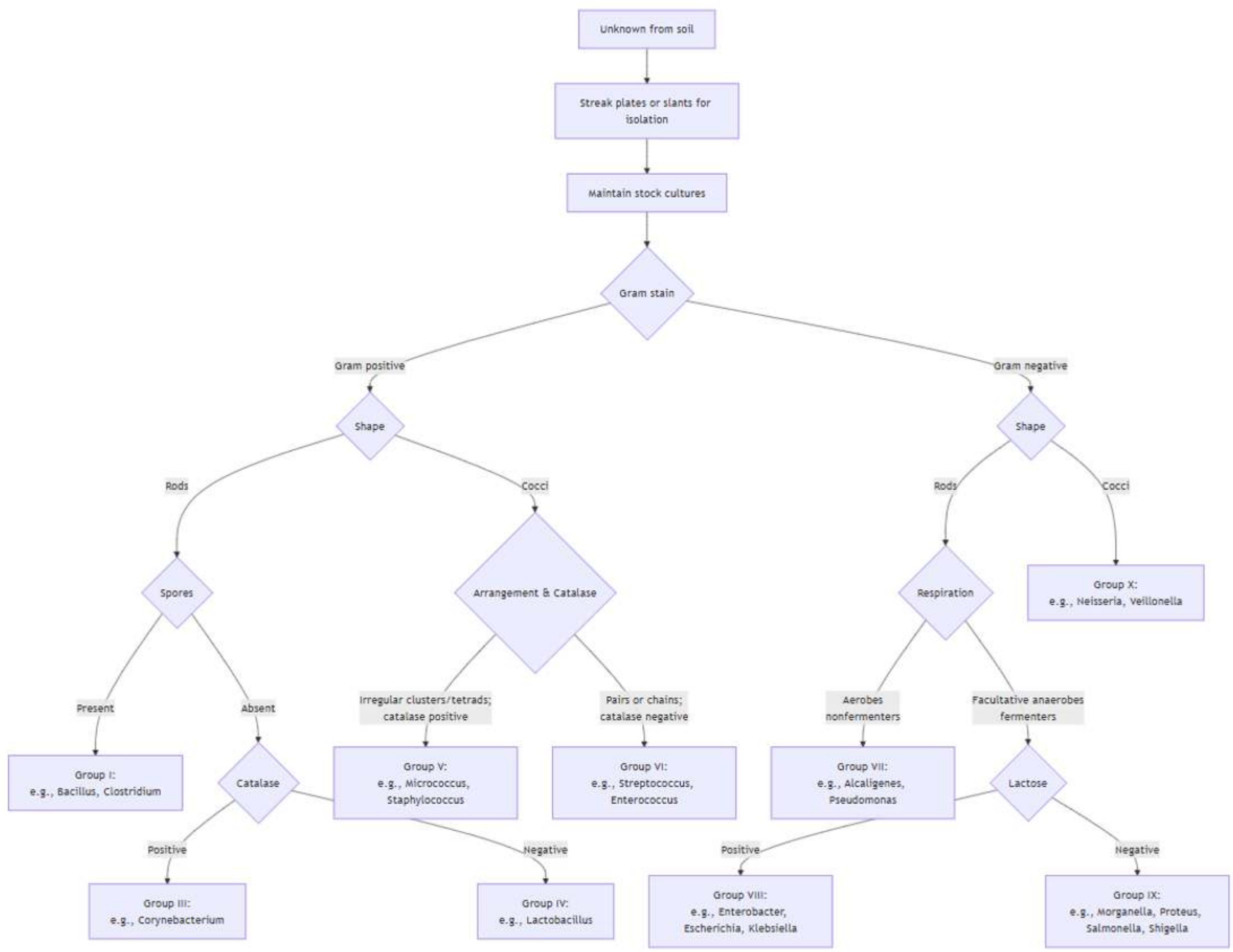
Record all raw data and observations here. Create a comprehensive table listing all tests performed, the results for known controls, and the results for your unknown isolate. Include observations like color changes, presence of bubbles/gas, clearing zones, or growth patterns.

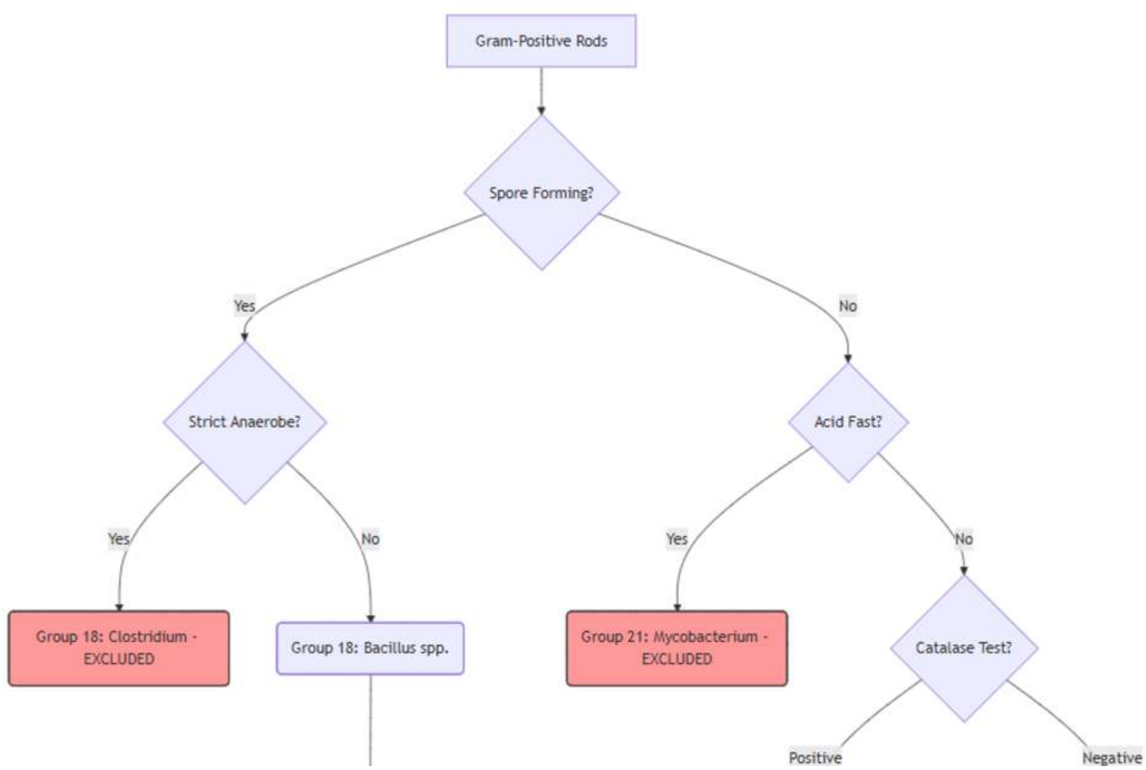
Conclusion

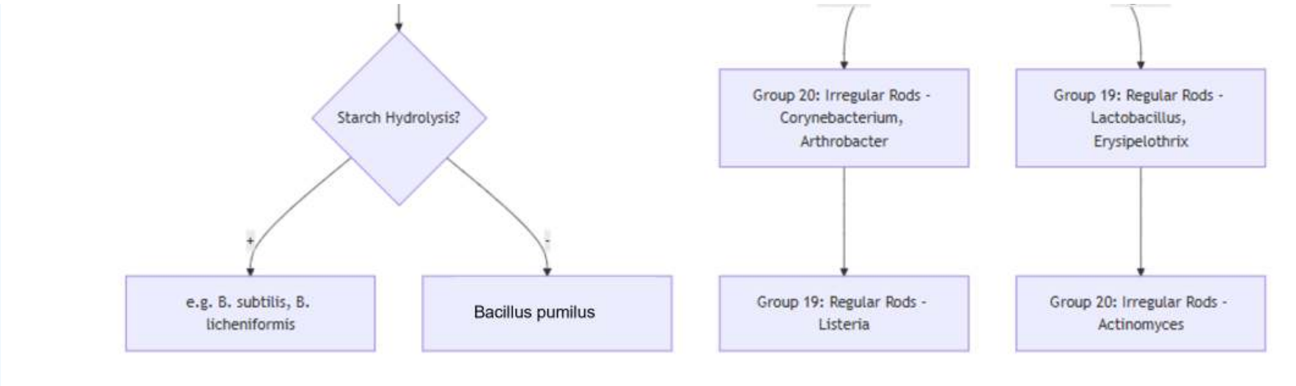
Write a summary that relates your results back to your hypothesis and purpose. State whether the hypothesis was supported or rejected. Discuss potential sources of error and explain how these results help narrow down the identification of your unknown. Be sure to name your identified bacteria. If you have sequencing results, do the biochemical results support your identification?

Important

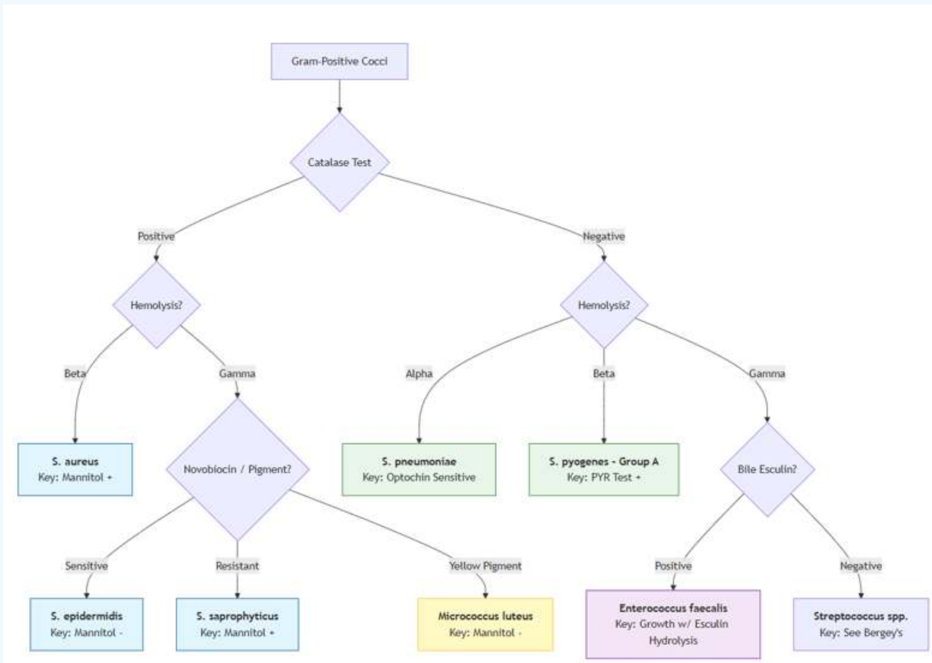
These sample flow charts are not all-encompassing, and other bacteria may have similar results. These charts allow you to determine what bacteria may be a "close relative" to your isolate. If you received DNA sequencing data, the bacteria identified with that may not match up with the limited number of bacteria presented in these charts.


Unknown from soil
Streak plates or slants for isolation
Maintain stock cultures
Gram stain
Gram positive
Gram negative
Shape
Rods
Cocci
Spores
Arrangement & Catalase
Respiration
Group X: e.g., Neisseria, Veillonella
Present
Absent
Irregular clusters/tetrads; catalase positive
Pairs or chains; catalase negative
Aerobes nonfermenters
Facultative anaerobes fermenters
Group I: e.g., Bacillus, Clostridium
Catalase
Group V: e.g., Micrococcus, Staphylococcus
Group VI: e.g., Streptococcus, Enterococcus
Group VII: e.g., Alcaligenes, Pseudomonas
Lactose
Positive
Negative
Group III: e.g., Corynebacterium
Group IV: e.g., Lactobacillus
Group VIII: e.g., Enterobacter, Escherichia, Klebsiella
Group IX: e.g., Morganella, Proteus, Salmonella, Shigella

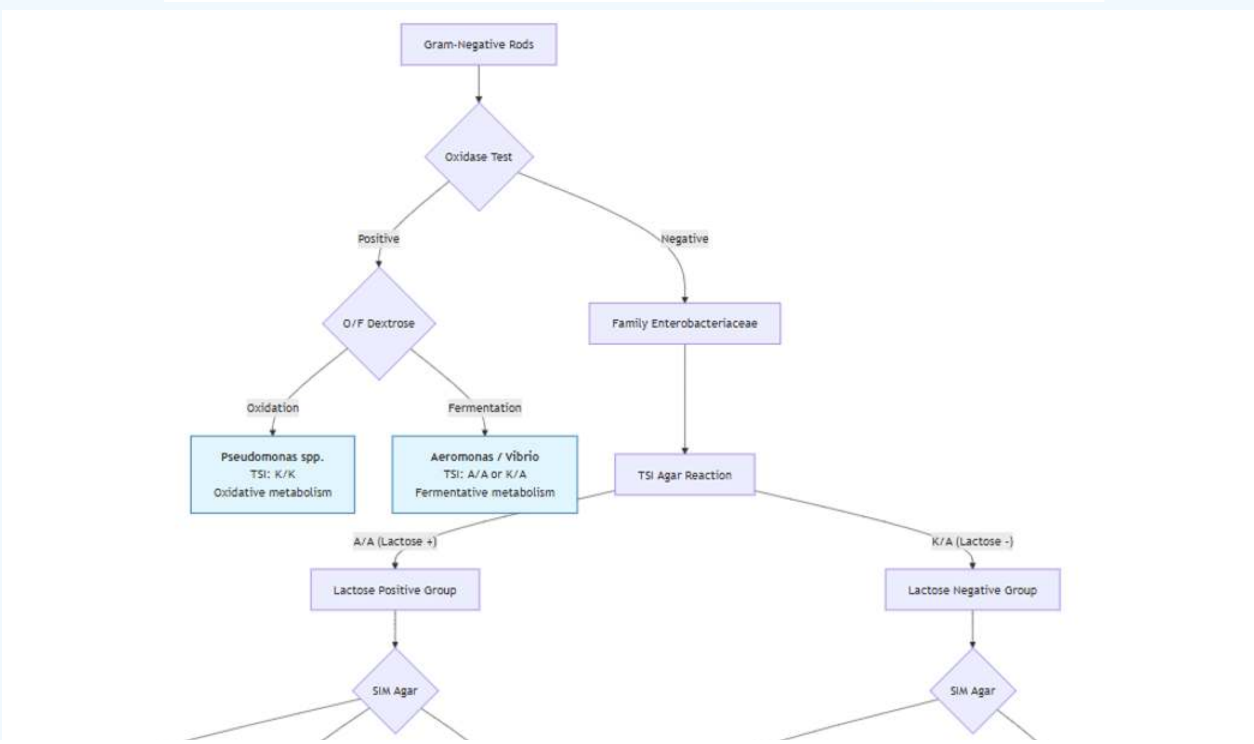

Gram-Positive Rods
Spore Forming?
Yes
No
Strict Anaerobe?
Acid Fast?
Group 18: Clostridium - EXCLUDED
Group 18: Bacillus spp.
Group 21: Mycobacterium - EXCLUDED
Catalase Test?
Positive
Negative



Starch Hydrolysis?
+
-
e.g. B. subtilis, B. licheniformis
Bacillus pumilus
Group 20: Irregular Rods - Corynebacterium, Arthrobacter
Group 19: Regular Rods - Lactobacillus, Erysipelothrix
Group 19: Regular Rods - Listeria
Group 20: Irregular Rods - Actinomyces



Gram-Positive Cocci
Catalase Test
Positive
Negative
Hemolysis?
Hemolysis?
Beta
Gamma
Alpha
Beta
Gamma
S. aureus
Key: Mannitol +
Novobiocin / Pigment?
S. pneumoniae
Key: Optochin Sensitive
S. pyogenes - Group A
Key: PYR Test +
Bile Esculin?
Sensitive
Resistant
Yellow Pigment
Positive
Negative
S. epidermidis
Key: Mannitol -
S. saprophyticus
Key: Mannitol +
Micrococcus luteus
Key: Mannitol -
Enterococcus faecalis
Key: Growth w/ Esculin Hydrolysis
Streptococcus spp.
Key: See Bergey's



Gram-Negative Rods
Oxidase Test
Positive
Negative
O/F Dextrose
Family Enterobacteriaceae
Oxidation
Fermentation
Pseudomonas spp.
TSI: K/K
Oxidative metabolism
Aeromonas / Vibrio
TSI: A/A or K/A
Fermentative metabolism
TSI Agar Reaction
A/A (Lactose +)
K/A (Lactose -)
Lactose Positive Group
Lactose Negative Group
SIM Agar
SIM Agar

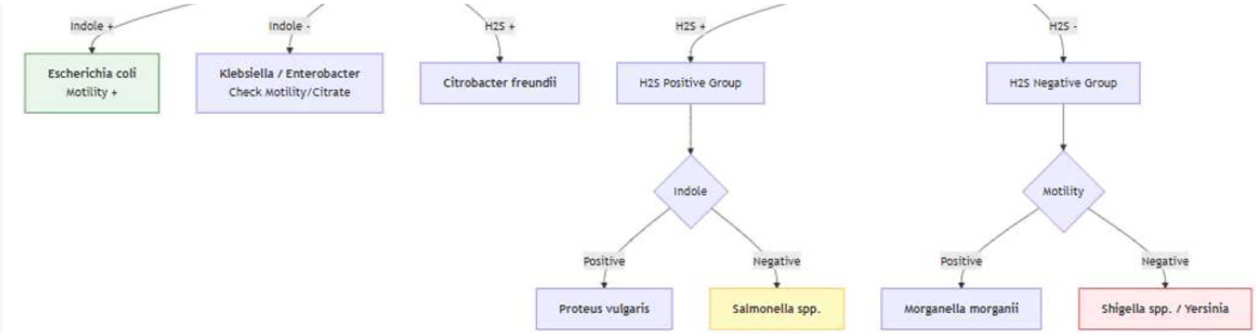


Figure 12.3.1: Sample Results Flowcharts (Created with Google Gemini)

Your Notebook Entry

CHAPTER OVERVIEW

Lab 13: Kirby-Bauer

13.1: Lab 13 Introduction

Kirby-Bauer and Skills Check

Learning Objectives

- Measure and interpret zones of inhibition to classify bacterial sensitivity to antibiotics.

Introduction

The Kirby-Bauer disk diffusion method is a standardized test used to determine a bacterium's susceptibility to antibiotics by measuring the zone of inhibition around an antibiotic disk on a Mueller-Hinton agar plate.

Materials

- Kirby-Bauer demo plates (pre-grown with Gram-positive or Gram-negative bacteria and 4 antibiotic disks)
- Calipers (for measuring zones in mm)

13.2: Lab 13 Protocol

Kirby-Bauer Test Interpretation

1. Obtain the pre-grown Kirby-Bauer demo plates.
2. Using calipers, measure the diameter (in millimeters) of the zone of inhibition for each antibiotic disk.
3. Record all measurements in your notebook.
4. Using the standardized interpretation chart provided by your instructor, classify the organism's sensitivity to each antibiotic as Susceptible (S), Intermediate (I), or Resistant (R).

13.3: Lab 13 Notebook Entry

Complete an entry with the following information for each section:

Notebook Entry

Purpose

Write 1-3 sentences here, defining the objective of interpreting antibiotic sensitivity results.

Hypothesis

Write an "If... then..." statement here, predicting the overall sensitivity of the demo bacteria.

Methods

Write a summary of the procedures used today for measuring zones of inhibition. Note any deviations from the manual. You may reference the manual and list only changes.

Results

Record all raw data and observations here. Use a table for the Kirby-Bauer test to list each antibiotic, its zone measurement (mm), and its interpretation (S/I/R).

Conclusion

Write a summary that relates your results back to your hypothesis and purpose. State whether the hypothesis was supported or rejected. Discuss potential sources of error and the importance of standardization in the Kirby-Bauer method.

Your Notebook Entry

CHAPTER OVERVIEW

14: Final Report and Submission

14.1: Final Report Introduction

Learning Objectives

- Consolidate and submit all lab notebook entries and the completed Isolate Identification Project (report in spring, poster in fall) report for grading.

Introduction

This final session is dedicated to reviewing the semester's material and submitting all completed work.

14.2: Final Report Protocol

Notebook Review

Review all entries in your lab notebook for completeness, adherence to the 5-section format (Purpose, Hypothesis, Methods, Results, Conclusion), and use of ink only.

Final Project Review

Your assignment is to summarize the work you've completed this semester in a report about the organism you identified. A detailed set of directions will be provided by your instructor. Be sure to include answers to the following:

1. What are you comparing about your soil samples, and why did you choose those sites to compare? What conclusions can be drawn about the number of antibiotic producers at each site?

These next questions will dive further into the microbes you identified using phenotypic and genotypic tests later in the semester.

2. What tests did you use to identify your microbe? Include an explanation of the meaning of your results and the degree of confidence you have in your identification.
3. According to sequence analysis, what was the closest relative (including accession number)?
4. What is known about the microbe you've identified? In what conditions is it known to flourish? Does it have any known connection to human health and/or disease? Does it possess known virulence factors? Are there any other interesting facts known about your microbe?

> **Important research:** Has this organism previously been known to secrete antibiotic chemicals? Against which of the ESKAPE pathogens is the antibiotic secreted bioactive?

Submission

Submit your complete lab notebook and your final report to your instructor as directed.

Cleanup

Dispose of all personal plates, tubes, tips, and other disposable materials into the appropriate biohazard bins or trash.

Dispose of your disposable lab coat in the trash as you leave the lab for the final time.

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